

FROM RETRIEVAL AUGMENTED GENERATION TO ANALOGY AUGMENTED GENERATION

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ABSTRACT

TODO

1 MOTIVATION

- Peng et al. (2023) demonstrate that augmenting with external knowledge sources and self-feedback improves performance on factual knowledge tasks.
- Huang et al. (2022) demonstrate that LLMs can produce procedural knowledge. But the VirtualHome environment is limited (and the choices of steps to include is subjective), and the paper does not leverage external knowledge (although it does leverage environment grounding).
- Factual knowledge is most succinctly represented by knowledge graphs, but these graphs leave out contextual information (which is why current approaches to factual generation use retrieved documents instead of KGs).
- Procedural knowledge can also be succinctly represented as a graph relating goals with steps, but this also leaves out contextual information.
- Retrieving procedural knowledge can be difficult because information is often spread across many more documents.
 - To reliably discover the information the model needs, it is already established practice to threshold retrieved documents based on embedding similarity with the query, or to prompt an LLM to filter or compress the passages.
- What is procedural knowledge: does pseudocode count? does instructions to get to the university count? what makes it procedural?
 - Procedural knowledge is action knowledge.
 - Procedural knowledge comes from *motivated* reasoning over factual knowledge.
- What we're doing is different from general summarization because the relevance of individual facts depends on the goal being achieved. It's also different from knowledge extraction because the resulting procedure has to cohesively combine many facts to produce steps, and the end result won't necessarily spell out the underlying facts.
- For performance on procedural knowledge, the constraint is procedural reasoning, not fact retrieval. (This is demonstrated by the performance of GPT-4 when just doing retrieval.)

2 LCSTEP

- talk about procedural knowledge design principles
- explain what a good procedure looks like (main idea steps, no unnecessary details, etc.) so others can follow (regardless of the schema they choose for their data)
- Our dataset is a procedural knowledge base for using langchain.
- The primary task will be procedure generation, and the downstream tasks will be code generation and (as a bonus) code writing Q&A.
- Each procedure also comes with unit tests that check whether generated code accomplishes the procedure.

LCStep contains three sets of documents: API reference, conceptual documentation, and procedures. See Figure 1 for a diagram of the process of generating the LCStep data.

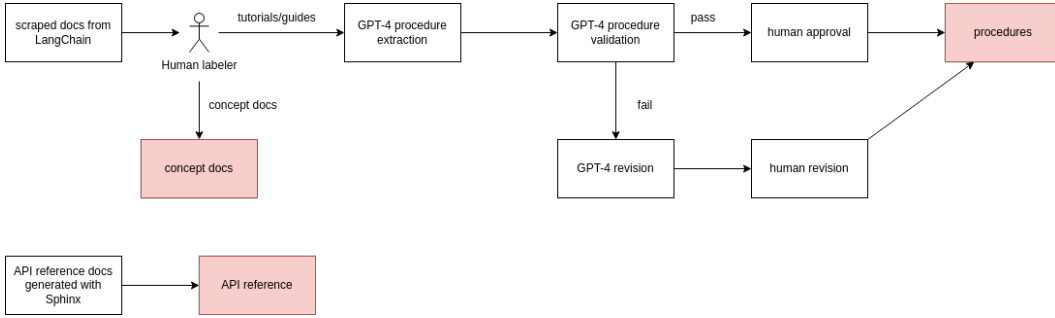


Figure 1: The workflow used to generate the LCStep dataset.

2.1 API REFERENCE

We generate the API reference material from the source files in the LangChain GitHub repository using Sphinx. These files contain descriptions of all APIs in the Python package, including call signatures and argument descriptions. These files do not contain any usage examples or high-level explanation.

2.2 CONCEPTUAL DOCUMENTATION AND PROCEDURES

We collect these documents by scraping the LangChain Python docs. We manually filter out topic pages and stubs, leaving 228 documents. We then manually classified these into around 30 documents of conceptual documentation, and around 180 documents containing tutorials/guides.

For the 180 tutorials/guides, we prompted GPT-4¹ to extract a list of high-level steps necessary to accomplish the goal. We then prompted GPT-4² to rate those extracted procedures using a list of criteria. We found that this caught many mistakes where GPT-4 did not follow all the stated instructions. In those cases, we had the model revise the steps to meet the requirements, and then we manually checked the revised versions.

3 ANALOGY-AUGMENTED GENERATION

The performance of retrieval-augmented generation (RAG) is highly dependent on the number and size of the documents, as well as the density of relevant information (citation needed). For many tasks, a lot of the knowledge to be retrieved is the same for many queries. In our case, many procedures involving the LangChain library require connecting to a language model, creating a prompt template, and constructing a chain. Instead of collecting this information anew for each similar query, we search over past outputs to find procedures similar to the current query, and then include them in the set of retrieved candidate documents to include in the final prompt.

- A good procedural knowledge representation provides context in natural language. This makes the individual parts composable.
- When we get a new task, we break it down into steps by comparing with similar tasks in the skill library.
- When we don't see a similar task or subtask, we retrieve additional documents to support compiling a new skill.
- This is like storing experience (i.e. how to do something) instead of facts (i.e. what something is) in memory.

¹See Appendix A.1 for the full prompt.

²See Appendix A.2 for the full prompt.

- This is like learning to use tools because we collect a library of “skills”, but it’s different because we don’t treat the skills as black boxes: we use them to draw analogies with to solve novel problems.³
- We can call this “analogy-augmented generation” (AAG).
- grounding:
 - This approach can still benefit from environment grounding much like Voyager or the two papers cited at the top.
 - In our case, we ground the output by transforming it into code and then evaluating the code. The first test is whether the code runs; the next is whether the LangChain model/chain/agent produces results that look right to a human or LLM, when compared with the examples from the tutorials.
 - This method works anywhere real or simulated feedback can be provided.

4 EXPERIMENTS

- understand how well plain RAG can produce procedural knowledge
 - model: GPT-4
 - compare model’s output on a few examples with and without the procedural knowledge base, by hand
 - highlight examples where procedural reasoning falls short with just with plain retrieval
 - test whether compressing the documents helps RAG (It’s likely that the standard model gets lost in long context and needs more compressed content.)
- full procedural knowledge generation test
 - models: GPT-3.5/4, Llama2 instruction-tuned (both prompted and fine-tuned [with PEFT?] versions of each) (plain, normal compression, and AAG versions)
 - evals:
 - * ROUGE
 - * self-check GPT
 - * GPT-4-rated relevancy, faithfulness, context recall, correctness
 - TODO borrow existing metrics, like OwlEval, or page 39 of FLASK
 - select a set of hardest procedures and report performance on that set
 - find examples where the model (un)successfully made analogy to produce the right procedure
 - Questions we want to answer with the RAG vs. fine-tuning experiments:
 - * How do we draw the line between using RAG and fine-tuning, based on the size and label types and publicness of the dataset?
- TODO experiment with token limits?
 - Set smaller token limits artificially for 100k models, and do the same for llama2 (the OSS standard).
- code generation
 - models: same as above, but we post-process the generated procedures to create code
 - evals:
 - * does the code run?
 - * “unit tests” collected from LCStep tutorials, as described in the methods section
- TODO code-writing question answering?

To demonstrate that AAG can overcome shortcomings of standard retrieval, we pit the two methods against each other on the task of generating the procedures in LCStep. We also test against a few baselines and oracles which we explain below. Both methods have access to the API reference and conceptual documentation from the dataset.

³See the Wikipedia articles for procedural knowledge and procedural memory.

4.1 BASELINES

All prompting methods are performed with GPT-4-0614 from OpenAI as well as an instruction-tuned version of LLaMA2.

- The **few-shot** approach will prompt the model with a basic instruction and a few fixed examples of the output format. We expect this method to fail due to a lack of knowledge of LangChain in the models’ training data.
- The vanilla retrieval-augmented approach (**RAG**) constructs a prompt by doing retrieval over a vector store containing the conceptual documentation and API reference material.
 - We expect to further refine this retrieval process somewhat. This may involve enforcing retrieval diversity by penalizing similarity between retrieved docs, having the model split the query into subqueries to find the individual parts necessary to build the whole answer, or using contextual compression.
 - This method would also benefit from HyDE (Gao et al., 2022), I³ (Dong et al., 2023), or EAR (Chuang et al., 2023). At that point, it might be worth having this optimized version as a separate baseline here.

For our experiment, the **AAG** approach constructs a prompt by retrieval over the same documents as RAG, plus all procedures generated so far. When retrieving from the past outputs, the model is separately presented with the selection prompt as described in Section ??.

4.2 EXTRA

- demo UI to demonstrate procedural knowledge-enhanced code copilot (more efficient? stronger reasoning? experience accumulation? better task solving strategy? better explainability?)

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A LCSTEP PROMPTS

A.1 INSTRUCTIONS FOR FORMATTING PROCEDURES

You are helping format procedures. I will give you raw procedure text, describing how to complete a procedure, with code and examples. Your task is to simplify and generalize this procedure to

achieve the desired format. The desired format is: a title, description, a minimal set of simple necessary instructions for general application where every instruction is describing one action and the API references to use in that action (with the module the API reference belongs to), and side notes if needed. Remove all unnecessary details, code, examples and specific cases. Keep only the strictly necessary steps to accomplish the procedure. The procedure should start with a title and a brief description containing any important information that doesn't fit into the instructions. Do not mention importing the required modules as a separate step, as we will include the required imports from the API references made in the instructions. If there are any checks, or case-specific instructions, they should be specified as a side note after the instructions as we want the procedure instructions to be for general use. If an API reference has parameters, make sure to mention these parameters in the instruction. If the raw text is actually describing multiple different procedures, you may then have the formatted procedure split into multiple different sets of instructions for each different procedure, but this should only be done if judged necessary.

A.2 INSTRUCTIONS FOR VALIDATING PROCEDURES

TODO